NO BLACK BOXES • PURE SIMPLE MATH

Let's Run Naive Bayes By Hand in 60 Seconds.

Forget scary Greek formulas. Today, you are the algorithm. We have **14 past days** of weather logs—and we need to predict if our team will play golf today.

🎯 TODAY'S MYSTERY WEATHER (INPUT X)

`X = [Sunny, Hot, Normal, False]`

OUTLOOK	TEMP	HUMIDITY	WINDY
Sunny	Hot	Normal	False

⛳ THE MODEL'S QUESTION

Will we Play Golf today or Stay Home?

YES ? NO ?

STEP 01
Split Past Days
Count YES vs NO days out of 14.

STEP 02
Score Both Teams
Multiply simple fractions for each clue.

STEP 03
Pick the Winner
Convert scores into final % odds.

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Swipe to see the 14 days →

BEFORE LOOKING AT TODAY'S WEATHER

Split Our 14 Past Days Into Two Buckets

Out of 14 recorded days, we played golf on **9 days** and stayed home on **5 days**. In ML, this starting baseline is called our **Prior Probability**.

BUCKET A • PLAYED GOLF

Team YES (9 Days)

P(Yes) = 9/14

Day	Outlook	Temp	Humidity	Windy
#01	Rainy	**Hot**	High	**False**
#03	Overcast	**Hot**	High	**False**
#05	**Sunny**	Cool	**Normal**	**False**
#07	Overcast	Cool	**Normal**	True
#09	Rainy	Cool	**Normal**	**False**
#10	**Sunny**	Mild	**Normal**	**False**
#11	Rainy	Mild	**Normal**	True
#12	Overcast	Mild	High	True
#13	Overcast	**Hot**	**Normal**	**False**

BUCKET B • STAYED HOME

Team NO (5 Days)

P(No) = 5/14

Day	Outlook	Temp	Humid	Wind
#02	Rainy	**Hot**	High	True
#04	**Sunny**	Mild	High	**False**
#06	**Sunny**	Cool	**Normal**	True
#08	Rainy	Mild	High	**False**
#14	**Sunny**	Mild	High	True

💡 **Notice the bold words!** Those match today's weather: *Sunny, Hot, Normal, False*. Next, we simply count them in each bucket!

🟢 **Starting Odds for YES: 9 / 14 (64.3%)**

🔴 **Starting Odds for NO: 5 / 14 (35.7%)**



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Swipe to score Team YES →

LOOKING ONLY INSIDE THE 9 "YES" DAYS

How Often Did Today's Weather Happen When We Played Golf?

Because we only look at the **9 YES days**, every single fraction below has **9 at the bottom**. Just count the green boxes!

☀️ Sunny	2 of 9 YES days were Sunny	2 / 9
🌡️ Hot Temp	3 of 9 YES days were Hot	3 / 9
💧 Normal Humid	6 of 9 YES days had Normal humidity	6 / 9
🍃 Wind = False	6 of 9 YES days were Calm (No wind)	6 / 9
⛳ Baseline YES	9 out of 14 total days in our dataset were YES days	9 / 14

MULTIPLY ALL 5 NUMBERS TOGETHER (NAIVE ASSUMPTION)

$(2/9) \times (3/9) \times (6/9) \times (6/9) \times (9/14)$

Raw YES Score

0.02116

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Now let's score Team NO →

LOOKING ONLY INSIDE THE 5 "NO" DAYS

How Often Did Today's Weather Happen When We Stayed Home?

Now switch buckets! Because there are only **5 NO days** in history, every clue fraction below has **5 at the bottom.**

☀️ **Sunny**	3 of 5 NO days were Sunny	3 / 5
🌡️ **Hot Temp**	Only 1 of 5 NO days was Hot	1 / 5
💧 **Normal Humid**	Only 1 of 5 NO days had Normal humidity	1 / 5
🍃 **Wind = False**	2 of 5 NO days were Calm (No wind)	2 / 5
🏠 **Baseline NO**	5 out of 14 total days in our dataset were NO days	5 / 14

MULTIPLY ALL 5 NUMBERS TOGETHER

$$(3/5) \times (1/5) \times (1/5) \times (2/5) \times (5/14)$$

Raw NO Score

0.00343

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Swipe for the Final Showdown →

TURNING RAW SCORES INTO 100% ODDS

Team YES Crushes Team NO by Over 6×!

Raw scores (`0.02116` vs `0.00343`) look tiny because we multiplied 5 fractions. To turn them into clean percentages, divide each score by their sum!

🏆 WINNER • PLAY GOLF

TEAM YES (PLAY GOLF)

86.1%

```
0.02116 ÷ (0.02116 + 0.00343)
```

Why it won: Normal humidity (6/9) and No wind (6/9) strongly favored playing golf.

TEAM NO (STAY HOME)

13.9%

```
0.00343 ÷ (0.02116 + 0.00343)
```

Why it lost: When we stayed home, humidity was almost always High—rarely Normal (1/5).

🤵 **Bonus Eagle-Eye Catch (If You Read the GeeksforGeeks Article!)**

In the 14-row table, **3 of 9 YES days** are Hot (3/9) and **1 of 5 NO days** is Hot (1/5), giving **86.1% YES vs 13.9% NO**. (The GFG text accidentally plugged in 2/9 and 2/5 for Hot to get `67.3%` vs `32.7%`—either way, **Play Golf = YES** wins!)

⛳ Final Prediction: PLAY GOLF (86.1%) 🏠 Stay Home (13.9%)

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1-Page Cheat Sheet & Python →

SAVE THIS SLIDE FOR INTERVIEWS

The Entire Algorithm in 3 Simple Rules + Python

Whether you are solving an exam problem on paper or running `scikit-learn` in production, Naive Bayes always follows these exact 3 steps:

RULE 01

Count Baselines

Find the starting fraction of each class in your dataset: `P(Yes)` and `P(No)`.

RULE 02

Multiply Clues

Inside each bucket, count how often each input feature appears and multiply them.

RULE 03

Normalize to 100%

Divide each team's score by the total sum of scores—highest percentage wins!

golf_naive_bayes.py • scikit-learn

```python
from sklearn.preprocessing import OrdinalEncoder
from sklearn.naive_bayes import CategoricalNB

# 1. Encode categorical weather columns & fit without smoothing (alpha=1e-10)
enc = OrdinalEncoder()
X_encoded = enc.fit_transform(df[['Outlook', 'Temp', 'Humidity', 'Windy']])
model = CategoricalNB(alpha=1e-10).fit(X_encoded, y)

# 2. Predict for Today: Sunny, Hot, Normal, False
today = enc.transform([['Sunny', 'Hot', 'Normal', 'False']])
print(model.predict_proba(today)) # Output: [[0.139, 0.861]] → 86.1% YES!
```

UP NEXT IN PART 04

Gaussian vs. Multinomial vs. Bernoulli — Which Naive Bayes Should You Pick?

Follow for Part 4 🔔

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Save & Repost 🚀